

Ans Sheet: CS & AI Master Exam - Devika Sengupta (Roll: CS-2026-004)

Q1: Backpropagation in Deep Neural Networks

Forward pass computes layer activations $z = Wx + b$ and prediction $\hat{y}_{\text{hat}}$.

Loss L is computed against target y using MSE or Cross-Entropy loss.

Backward pass computes gradients via calculus chain-rule:

$$dL/dW = (dL/da) * (da/dz) * (dz/dW) \text{ where } dz/dW = x.$$

Weight update: $W = W - \text{eta} * (dL/dW)$. Gradients flow back layer by layer.

Q2: Time & Space Complexity: QuickSort vs MergeSort

1. MergeSort: Divide & conquer algorithm.

- Best/Avg/Worst Time: $O(N \log N)$ in all cases.
- Space: $O(N)$ auxiliary buffer for merging subarrays.

2. QuickSort: Partitioning around chosen pivot element.

- Average Time: $O(N \log N)$.

- Worst ~~Time: $O(N^2 \log N)$~~

^ Worst Time: $O(N^2)$ when array sorted and bad pivot chosen!

- Space: $O(\log N)$ recursion call stack (in-place).

Q3: ACID Properties in DBMS

- Atomicity: All or nothing transaction. If query fails, rollback completely.

- Consistency: Database transitions between valid states adhering to FK schema.

- Isolation: Concurrent sessions run independently without dirty reads.

- Durability: Committed updates written to non-volatile WAL log.

Q4: Object-Oriented Programming (OOP) Principles

1. Encapsulation: Bundling data and methods, private fields with getters/setters.

2. Abstraction: Hiding implementation details via interfaces & abstract classes.

3. Inheritance: Subclass inherits behavior and fields from superclass.

4. Polymorphism: Method overriding at runtime and overloading at compile-time.

Q5: RSA Public Key Cryptography

Select large primes p, q . Modulus $n = p \cdot q$. Euler $\phi(n) = (p-1)(q-1)$.

Public key exponent e coprime to $\phi(n)$.

Private key d : $e \cdot d \equiv 1 \pmod{\phi(n)}$

$$\wedge e \cdot d \equiv 1 \pmod{\phi(n)}$$

Encryption: $c = m^e \pmod{n}$ (using public key (e, n)).

Decryption: $m = c^d \pmod{n}$ (using private key (d, n)).